

Submit your written answers as a PDF to both **Crowdmark** and **LEARN Dropbox**. For the coding exercises, also submit your code as **a1-code.zip** to LEARN, with clearly named scripts and a README giving the commands used to obtain your results. Do not include the data, virtual environment, or model checkpoints. Work individually and follow the course academic-integrity and generative-AI policies, including disclosure of permitted assistance. LaTeX is optional.

[16 marks] Question 1: Coding exercises

Use the supplied Adult dataset and starter code. The target Y is recorded annual income: $Y = 1$ means income above US\$50,000, and $Y = 0$ means income at or below that threshold. The group attribute A uses the dataset's recorded sex categories: $A = 0$ is **Female** and $A = 1$ is **Male**.

The original training and test sets contain 32,561 and 16,281 examples, respectively. The loader removes **sex_Female** and **sex_Male** from the inputs, leaving 111 features, and keeps A separately. The final header entry, **income**, is the target name, not an input feature. Standardization uses training-set statistics only. Use **raw_x** for correlations and **x** for model inputs.

Preliminaries

Let $\hat{Y}_i \in \{0, 1\}$ be the classifier's prediction on example i . For n evaluation examples, let n_a be the number with $A_i = a$. We use three metrics:

$$\begin{aligned}\text{Acc} &= \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\hat{Y}_i = Y_i\}, \\ \text{RWA}_{\text{acc}_Y} &= \frac{1}{2} \sum_{a \in \{0,1\}} \frac{1}{n_a} \sum_{i:A_i=a} \mathbf{1}\{\hat{Y}_i = Y_i\}, \\ \Delta_{\text{DP}} &= \left| \frac{1}{n_0} \sum_{i:A_i=0} \hat{Y}_i - \frac{1}{n_1} \sum_{i:A_i=1} \hat{Y}_i \right|.\end{aligned}$$

Accuracy measures how often the classifier is correct. Reweighted accuracy weights the two groups equally. Δ_{DP} is the absolute difference between their positive-prediction rates. Compute these metrics over the entire evaluation set; do not average batch-level rates or gaps. Treat a rate with a zero denominator as undefined.

For a classifier h predicting A instead of Y , define its population reweighted accuracy as

$$\text{RWA}_{\text{acc}_A}(h) = \frac{1}{2} [P(h(X) = 0 \mid A = 0) + P(h(X) = 1 \mid A = 1)].$$

[3 marks] Q1.1: Feature correlations

On the **training set**, identify the ten features most correlated with Y and the ten most correlated with A , ranked by absolute Pearson correlation. Report feature names and signed correlation values. Ignore undefined correlations; break ties by feature-column order. Implement `correlations` in `student.py`.

[4 marks] Q1.2: Classifier evaluation

Implement overall accuracy, reweighted accuracy, and Δ_{DP} in `student.metrics`. Train the supplied classifier on the training set, and report these three metrics on the **test set**. Which group has the higher average binary prediction $\hat{Y}$? Give the positive-prediction rate for each group.

The supplied model has one hidden layer with 64 ReLU units and one output logit. Use the supplied training defaults: binary cross-entropy, Adam with learning rate 10^{-3} and weight decay 10^{-5} , batch size 256, and 10 epochs. The binary prediction is $\hat{Y} = \mathbf{1}\{r(X) \geq 0.5\}$, where $r(X)$ is the sigmoid score. No GPU is required.

[2 marks] Q1.3: Correlations with predictions

Using the classifier from Question 1.2 and the **test set**, identify the three features most correlated with $\hat{Y}$:

- across all test examples;
- among examples with $A = 0$; and
- among examples with $A = 1$.

Rank by absolute Pearson correlation and report signed values. Handle undefined correlations and ties as in Question 1.1.

[7 marks] Q1.4: Demographic-parity penalty

Modify the training objective to include a differentiable term estimating the lack of demographic parity, with a hyperparameter α controlling its weight relative to the classification loss. Implement this term in `student.dp_penalty`.

(a) [3 marks] State the mathematical form of your regularizer and discuss its design, including why it is differentiable with respect to the network parameters. A function of hard predictions obtained by thresholding the network output does not provide a suitable training gradient. You may instead use the continuous scores. The supplied loop skips the penalty on a batch missing either group, using the classification loss alone.

(b) [4 marks] Determine and report a range of α that produces useful comparisons. Train a sequence of classifiers over this range, including the unregularized baseline $\alpha = 0$. Plot **test-set** overall accuracy, reweighted accuracy, and Δ_{DP} as functions of α . Use at least five values including zero, and keep the training settings and random seed fixed across runs. Briefly describe the accuracy–DP tradeoff in your plot.

Choose the range using training runs, then evaluate all chosen values on the test set. This exercise compares the resulting curves; it does not require selecting one model or establishing that a particular model is best. You are not required to obtain a monotone trend.

[24 marks] Question 2: Written exercises

Give explicit constructions and show the probability calculations or proof. These questions concern population distributions, not estimates from Adult.

[6 marks] Q2.1: Conditional independence

Construct three finite-valued random variables X , R , and Y such that $X \perp R$ but $X \not\perp R \mid Y$. Specify their joint distribution and verify both claims. Here R is an arbitrary random variable; it need not be a function of X .

[8 marks] Q2.2: A three-valued target

Give an example of a classification problem where the target Y takes three distinct values, and independence and separation are simultaneously achievable, in the case where A and Y are not independent.

Specify a joint distribution of group membership A , target Y , and prediction R . Verify that

$$R \perp A, \quad R \perp A \mid Y, \quad A \not\perp Y.$$

Both groups and all three target values must have positive probability.

[10 marks] Q2.3: Predicting group membership

Suppose you have trained a binary classifier g to predict Y from X . Assume $P(A = 0) > 0$ and $P(A = 1) > 0$, and define its population DP gap by

$$\Delta_{\text{DP}}(g) = |P(g(X) = 1 \mid A = 1) - P(g(X) = 1 \mid A = 0)|.$$

Show that there exists a classifier h predicting A from X with

$$\text{RWAcc}_A(h) \geq \frac{1}{2} \Delta_{\text{DP}}(g) + \frac{1}{2}.$$

Construct h and prove the bound, including both signs of the difference inside the absolute value.